

Torque Specification TS-100 - Packers, Bridge Plugs, Cement Retainers & Frac Plugs

Engineering Specification TS-100 Rev D | Effective 2025-09-15

1. Purpose and Conditions

This specification defines make-up torque for threaded connections on ACT packer and plug products. Values assume clean, undamaged threads with API 5A3 modified thread compound (friction factor 1.0) applied to pin threads only. For other compounds multiply by the manufacturer's friction factor. Torque equipment must carry calibration within 12 months.

2. Table 1 - API Tubing Connections (Tool Body End Connections)

Connection	Size	Min (ft-lb)	Optimum (ft-lb)	Max (ft-lb)
EUE 8RD	2-3/8 in.	1,150	1,290	1,430
EUE 8RD	2-7/8 in.	1,650	1,850	2,050
EUE 8RD	3-1/2 in.	2,280	2,560	2,840
NUE 10RD	2-3/8 in.	870	970	1,070
NUE 10RD	2-7/8 in.	1,240	1,380	1,520
LTC casing	4-1/2 in.	3,000	3,400	3,800

3. Table 2 - Packer Body Connections (Shouldered)

Product	Joint	Min (ft-lb)	Optimum (ft-lb)	Max (ft-lb)
ACT-PKR-4500	Top/bottom sub to mandrel	2,800	3,000	3,200
ACT-PKR-5500	Top sub to mandrel	2,800	3,000	3,200
ACT-PKR-5500	Guide to mandrel	1,900	2,100	2,300
ACT-PKR-7000	Top/bottom sub to mandrel	5,200	5,600	6,000
ACT-CR-5500	Valve body to mandrel	900	1,000	1,100

4. Table 3 - Setting Adapter Kits

Kit	Joint	Min (ft-lb)	Optimum (ft-lb)	Max (ft-lb)
ACT-BP-AK-45	Tension rod to crossover	80	90	100
ACT-BP-AK-55	Tension rod to crossover	80	90	100
ACT-FP-AK-55	Tension rod to setting tool	80	90	100
All kits	Setting sleeve to sleeve adapter	150	165	180

5. Table 4 - Set Screws and Fasteners (values in ft-lb unless noted)

Fastener	Size / Grade	Dry	Lubricated	Notes
Slip retainer screw (PKR-4500)	1/4-20 UNC Gr 8	11	10	Witness mark after torque
Drag block screw (PKR-4500)	3/8-16 UNC Gr 8	27	25	-
Slip retainer screw (PKR-7000)	5/16-18 Inconel 718	20	18	Anti-seize mandatory
Body lock screw (SSD-2875)	1/4-20 Gr 8	17	15	Thread locker medium
Shear screws (all brass)	10-24 / 1/4-28	SNUG	finger + 1/4 turn	NEVER torque to yield

6. General Notes

- Shear screws and shear pins are NEVER torqued to a wrench value - install finger tight plus 1/4 turn and witness mark.
- Premium connections: use licensor's published torque; Table 1 does not apply.
- Reject any connection that reaches max torque more than 1/8 turn before shoulder contact.
- Reference documents: API RP 5C1, API 5B thread gauging, ACT drawing notes.

Role	Name	Signature	Date
Prepared (Mfg Engineering)	T. Okafor	/signed electronically/	2025-09-15
Approved (Engineering Manager)	J. Martinez, P.E.	/signed electronically/	2025-09-15
Document Control	L. Hargrove	/signed electronically/	2025-09-15